Variance-Reduction Composition in Conditional Flow Matching: From 2D Experiments to CIFAR-10

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Abstract

Conditional Flow Matching (CFM) enables simulation-free training of continuous normalizing flows by regressing a neural velocity field against a target conditional vector field. However, the single-sample conditional target exhibits high gradient variance, which degrades sample quality and slows convergence. Three independent lines of work address this variance through different axes: *coupling* (replacing independent noise–data pairing with minibatch optimal transport), *objective* (approximating the marginal vector field via importance-weighted multi-reference targets), and *timestep estimation* (antithetic temporal sampling with consistency regularization). Whether these techniques compose additively or interfere remains unexplored.

We investigate this question through two complementary studies. **Study 1** compares four flow-matching variants (OT-CFM, VP-CFM, Target-CFM, SB-CFM) on 2D synthetic distributions, confirming that OT-type paths produce straighter trajectories and better low-budget robustness, while Schrödinger Bridge paths achieve the best endpoint quality. **Study 2** crosses the three variance-reduction axes in a full 2^3 factorial ablation on CIFAR-10 (8 cells, U-Net backbone, 80k training steps). Only one cell converged: BatchOT + CFM + Uniform (cell_100), achieving FID 18.4. The remaining seven cells diverged to NaN or exhibited severely unstable training.

These negative results are informative: they reveal that StableVM's importance weights scale catastrophically with data dimension (d=3072), that TPC's consistency penalty introduces gradient explosions at complementary timesteps, and that BatchOT coupling is the single most impactful variance-reduction technique. The three axes are not additive—combining them is destructive in high dimensions.

1 Introduction

Generative modeling has undergone a paradigm shift with the success of diffusion models (**??**), which learn to reverse a noise-corruption process to generate samples. While highly effective, these models require simulating stochastic differential equations (SDEs) during both training and sampling, leading to computational overhead and sensitivity to discretization.

Flow Matching (FM), introduced by **?**, offers an alternative based on continuous normalizing flows (CNFs). Instead of learning to reverse a diffusion process, FM directly regresses a neural velocity field $v_\theta(x, t)$ against a target vector field that transports a simple prior p_0 (e.g., standard Gaussian) to the data distribution p_1. The key advantage is simulation-free training: no ODE must be solved during the training loop, only at sampling time.

Conditional Flow Matching (CFM) makes FM practical by decomposing the intractable marginal vector field into tractable conditional terms. Each training sample defines a conditional probability path from noise to data, and the velocity field is trained to match the conditional vector field along this path. This yields a simple regression loss that is unbiased for the marginal FM objective.

The variance problem. While unbiased, the CFM loss has high variance because each training step sees only a single conditional vector field sample. This variance manifests as noisy gradients, slower convergence, and degraded sample quality at low numbers of function evaluations (NFE). Three independent research directions have proposed solutions along different axes:

1. **Coupling** — replacing independent noise–data pairing ($x_0 \sim p_0$ independent of $x_1 \sim p_1$) with minibatch optimal transport (BatchOT), which finds low-cost pairings within each minibatch (**?**). This straightens trajectories and reduces the magnitude of the velocity field.
2. **Objective** — replacing the single-sample conditional target with an importance-weighted average over multiple reference samples, approximating the marginal vector field more accurately (StableVM).
3. **Timestep estimation** — replacing uniform timestep sampling with antithetic temporal pairs and a consistency penalty that couples velocity predictions at complementary timesteps (TPC).

Research questions. Each technique has been shown effective in isolation, but their joint behavior is unknown. We ask:

RQ1 Are the three variance-reduction axes additive when composed?

RQ2 Which single axis provides the largest improvement?

RQ3 Do insights from 2D synthetic experiments transfer to high-dimensional image generation?

Approach. We address these questions through two complementary studies. Study 1 (Section 4) compares four flow-matching path variants on 2D distributions, providing geometric intuition about trajectory straightness, solver robustness, and sample quality. Study 2 (Section 5) crosses the three variance-reduction axes in a full $2^3 = 8$-cell factorial ablation on CIFAR-10, measuring FID, training stability, trajectory geometry, and per-timestep variance.

Preview of findings. The results are striking and largely negative for the composition hypothesis. Only one of eight CIFAR-10 cells converged (BatchOT + CFM + Uniform, FID 18.4). The remaining seven diverged or oscillated. We provide detailed failure analysis showing that the techniques interfere catastrophically in high dimensions due to numerical instabilities that are invisible in 2D. BatchOT emerges as the single most impactful technique, consistent with **?**.

2 Background

2.1 Continuous Normalizing Flows

A continuous normalizing flow (CNF) defines a time-dependent diffeomorphism $\phi_t : \mathbb{R}^d \to \mathbb{R}^d$ via an ordinary differential equation (ODE):

$$\frac{d\phi_t(x)}{dt} = v_\theta(\phi_t(x), t), \quad \phi_0(x) = x, \tag{1}$$

where $v_\theta : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$ is a neural velocity field parameterized by θ. If p_0 denotes a simple prior (e.g., $\mathcal{N}(0, I)$), then $p_t = [\phi_t]_\# p_0$ is the pushforward distribution at time t, and the goal is $p_1 \approx q$ where q is the data distribution.

2.2 Flow Matching

? propose to train v_θ by directly regressing against a target vector field u_t that generates a prescribed probability path p_t from p_0 to p_1. The **Flow Matching** loss is:

$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}[0,1]}\mathbb{E}_{x\sim p_t}\left\|v_\theta(x,t) - u_t(x)\right\|^2. \tag{2}$$

This loss is intractable because p_t and u_t depend on the full data distribution. The key insight is to decompose via conditional probability paths.

2.3 Conditional Flow Matching

Define a *conditional* probability path $p_t(x|x_1)$ for each data point $x_1 \sim q$, with corresponding conditional vector field $u_t(x|x_1)$. The marginal path is recovered as $p_t(x) = \int p_t(x|x_1)q(x_1)\,dx_1$. The **CFM loss** is:

$$\mathcal{L}_{\mathrm{CFM}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}[0,1]}\mathbb{E}_{x_1\sim q}\mathbb{E}_{x\sim p_t(\cdot|x_1)}\left\|v_\theta(x,t) - u_t(x|x_1)\right\|^2. \tag{3}$$

? prove (Theorem 2) that $\nabla_\theta\mathcal{L}_{\mathrm{CFM}} = \nabla_\theta\mathcal{L}_{\mathrm{FM}}$, so minimizing the tractable CFM loss is equivalent to minimizing the FM loss.

2.4 OT Probability Path

The **optimal-transport** conditional path (Eq. 22 of **?**) uses linear interpolation:

$$\mu_t(x_1) = t\cdot x_1, \tag{4}$$

$$\sigma_t = 1 - (1-\sigma_{\min})t, \quad \sigma_{\min} = 10^{-4}, \tag{5}$$

$$p_t(x|x_1) = \mathcal{N}\big(x \,\big|\, \mu_t(x_1),\, \sigma_t^2 I\big), \tag{6}$$

with conditional vector field:

$$u_t(x|x_1) = \frac{x_1 - (1-\sigma_{\min})x}{1-(1-\sigma_{\min})t}. \tag{7}$$

The sampling process interpolates $x_t = \mu_t(x_1) + \sigma_t\, x_0$ where $x_0 \sim \mathcal{N}(0, I)$.

2.5 BatchOT Coupling

In standard CFM, the noise $x_0 \sim p_0$ is sampled independently of $x_1 \sim q$. **?** propose **minibatch optimal transport coupling**: within each minibatch of size B, solve an entropic OT problem between the B noise samples and B data samples to find an assignment that minimizes total squared Euclidean cost. The implementation uses:

1. Sinkhorn's algorithm (**?**) in log-domain with regularization $\epsilon = 0.05$ and 100 iterations, yielding an approximate transport plan $\pi \in \mathbb{R}_+^{B\times B}$.

2. The Hungarian algorithm to extract a deterministic one-to-one assignment from π, respecting both marginals exactly.

The resulting paired samples $(x_0^{\pi(i)}, x_1^i)$ produce shorter, straighter interpolation paths, reducing variance of the conditional vector field.

2.6 StableVM Objective

Instead of regressing against the single-sample conditional target $u_t(x|x_1)$, the StableVM objective approximates the *marginal* vector field using K reference samples from a FIFO memory bank. For each training point x_t at time t:

$$w_k = \mathrm{softmax}_k \left(\log p_t(x_t|x_1^{(k)}) \right), \quad k = 1, \ldots, K, \tag{8}$$

$$u_t^{\mathrm{VM}}(x_t) = \sum_{k=1}^{K} w_k \cdot u_t(x_t|x_1^{(k)}), \tag{9}$$

where $\{x_1^{(k)}\}_{k=1}^K$ are drawn from the memory bank. The log-probability under the OT path is:

$$\log p_t(x|x_1) = -\frac{d}{2}\log(2\pi) - d\log\sigma_t - \frac{1}{2\sigma_t^2}\|x - \mu_t(x_1)\|^2, \tag{10}$$

where d is the data dimension. The term $-d\log\sigma_t$ will prove critical in our failure analysis (Section 6).

2.7 TPC Estimator

The Temporal Pair Consistency (TPC) estimator replaces uniform timestep sampling with antithetic pairs:

$$t_1 \sim \mathcal{U}[0, 0.5], \quad t_2 = 1 - t_1, \tag{11}$$

and adds a consistency penalty coupling velocity predictions at complementary timesteps:

$$\mathcal{L}_{\mathrm{TPC}} = \mathcal{L}_{\mathrm{CFM}}(t_1) + \mathcal{L}_{\mathrm{CFM}}(t_2) + \lambda \left\| v_\theta(x_t^{(1)}, t_1) - v_\theta(x_t^{(2)}, t_2) \right\|^2, \tag{12}$$

where $\lambda = 0.1$ and $x_t^{(i)}$ are freshly sampled at each timestep. The antithetic design ensures coverage of both time extremes in every step, and the penalty encourages temporal coherence.

3 Methodology

3.1 Study 1: 2D Synthetic Experiments

Datasets. Three 2D synthetic distributions are used: `checkerboard` (8 modes in a grid pattern), `two_moons` (two interleaving crescents), and `gaussian_mixture` (a mixture of Gaussians). These distributions provide direct visual access to generated samples and transport trajectories.

Variants compared. Four flow-matching variants are trained with identical architectures:

- **OT-CFM**: Optimal-transport conditional path (Section 2, Eq. 4–7).
- **VP-CFM**: VP-diffusion conditional path, using the variance-preserving noise schedule of **?** with $\beta_{\min} = 0.1$, $\beta_{\max} = 20$.
- **Target-CFM**: A path conditioned on both source and target, without OT coupling.
- **SB-CFM**: Schrödinger Bridge CFM, using the entropic OT formulation.

Architecture. The velocity field is a time-conditioned MLP with 3 hidden layers of 256 units and SiLU activations. Training uses Adam (**?**) with learning rate 10^{-3} for 10,000 steps.

Evaluation metrics.

- **SWD** (Sliced Wasserstein Distance): Endpoint sample quality, computed between 10,000 generated and reference samples.
- **MMD** (Maximum Mean Discrepancy): Second endpoint quality metric with Gaussian kernel.
- **NFE-vs-SWD curves**: Sample quality as a function of the number of ODE solver function evaluations, testing robustness to limited compute budgets.
- **Trajectory curvature**: Mean second-order finite difference along sampled ODE trajectories; lower values indicate straighter paths.
- **Time-to-structure AUC**: Area under the SWD-over-time curve, measuring how quickly meaningful structure appears during the ODE integration.

3.2 Study 2: CIFAR-10 Factorial Ablation

Experimental design. The three variance-reduction axes are crossed in a full 2^3 factorial design, yielding 8 cells. Table 1 summarizes the design.

Table 1: 2^3 factorial ablation design. Binary encoding: bit 2 = Coupling, bit 1 = Objective, bit 0 = Estimator. A value of 0 selects the baseline; 1 selects the improved variant.

Cell	Binary	Coupling	Objective	Estimator
0	000	Independent	CFM	Uniform
1	001	Independent	CFM	TPC
2	010	Independent	StableVM	Uniform
3	011	Independent	StableVM	TPC
4	100	BatchOT	CFM	Uniform
5	101	BatchOT	CFM	TPC
6	110	BatchOT	StableVM	Uniform
7	111	BatchOT	StableVM	TPC

Architecture. All cells share a U-Net backbone with:

- Base channels: 64 (fast mode), channel multipliers $[1, 2, 2, 2]$.
- Sinusoidal time embedding (dimension 64) projected through a 2-layer MLP with SiLU.
- Multi-head self-attention (4 heads) at 16×16 resolution.
- 2 residual blocks per resolution level, with GroupNorm(32) and dropout.
- Total: approximately 9M parameters.

Training. All cells are trained with identical hyperparameters:

- Optimizer: Adam (lr $= 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$).
- EMA decay: 0.9999. Mixed precision: fp16.
- Batch size: 256. Total steps: 80,000 (20.5M images seen).
- Dataset: CIFAR-10 training set (50,000 images, $32 \times 32 \times 3$).
- Hardware: AMD MI210 GPUs (ROCm 6.0) on a SLURM cluster.

Diagnostics. Every 5,000 steps, three diagnostic quantities are computed on a held-out batch of 64 samples:

- **Curvature proxy**: Mean second-order finite difference along Euler trajectories (50 NFE).
- **Backtracking fraction**: Proportion of intermediate steps where the velocity points away from the final endpoint (cosine angle < 0).
- **Per-timestep variance**: Variance of per-sample squared errors at 20 evenly-spaced timesteps in $[0.05, 0.95]$.

Evaluation. After training, each cell is evaluated by generating 10,000 samples at NFE $\in \{5, 10, 20, 40, 100\}$ with three ODE solvers (Euler, midpoint, RK4). FID (**?**) is computed via `torch-fidelity` against the CIFAR-10 training set. This yields 15 FID measurements per cell.

Checkpoint selection. For cells that diverged to NaN during training (cells 000, 001, 101), the checkpoint at step 10,000 (before divergence) was used for evaluation. All other cells used the final checkpoint. This policy ensures that every cell produces FID numbers, even if they reflect a partially-trained model.

4 Study 1: 2D Synthetic Experiments

This section presents results from comparing four flow-matching path variants on 2D synthetic distributions. The goal is to build geometric intuition about how path design affects trajectory quality, solver robustness, and sample fidelity.

4.1 Single-Dataset Analysis: Checkerboard

Table 2: Endpoint quality on the checkerboard distribution at NFE=64. Lower is better for both metrics.

Variant	SWD@64	MMD@64
OT-CFM	0.02049	0.000503
VP-CFM	0.02655	0.000792
Target-CFM	0.03004	0.001068
SB-CFM	**0.02016**	**0.000449**

SB-CFM achieves the best endpoint quality, improving SWD by 24% and MMD by 43% over VP-CFM. OT-CFM is a close second, within 2% of SB-CFM on SWD. Figure 1 shows generated samples from all four variants.

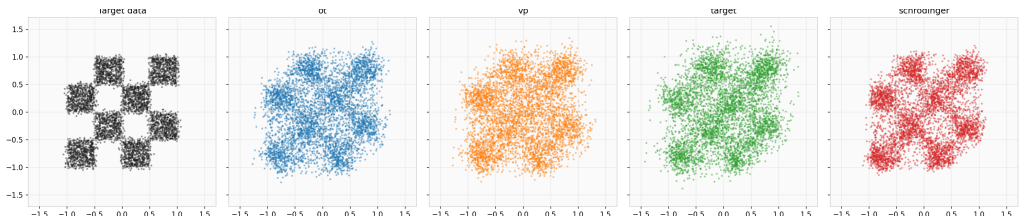


Figure 1: Generated samples on the checkerboard distribution at NFE=64. All variants capture the mode structure, but VP-CFM and Target-CFM show more dispersion between modes.

4.2 NFE Robustness

The low-budget regime is more discriminative than endpoint quality. Figure 2 shows SWD as a function of NFE for all four variants.

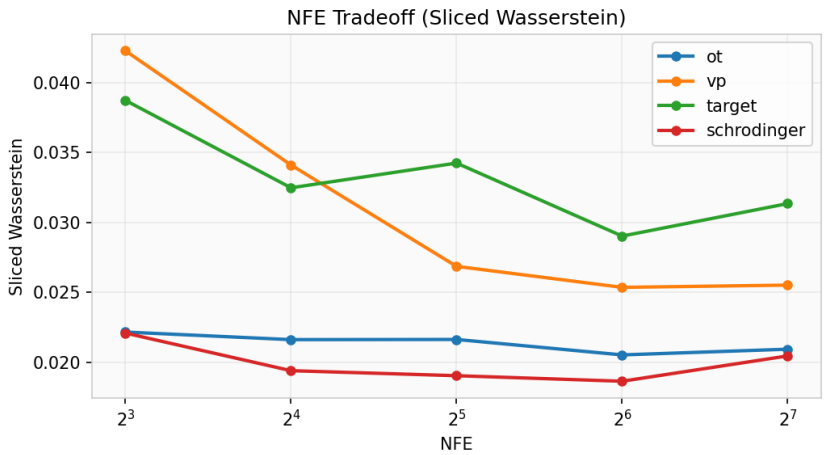


Figure 2: SWD vs. NFE on the checkerboard distribution. OT-CFM and SB-CFM maintain quality at low NFE, while VP-CFM degrades significantly below NFE=16.

At low NFE (≤ 8), OT-CFM and SB-CFM achieve SWD around 0.022, while VP-CFM and Target-CFM reach approximately 0.042 and 0.039 respectively—roughly 47% worse. As NFE increases, VP-CFM improves strongly, indicating high discretization sensitivity. OT-type paths are inherently more solver-robust because their straighter trajectories require fewer integration steps to approximate accurately.

4.3 Trajectory Geometry

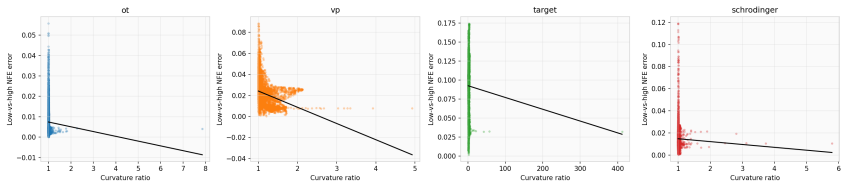


Figure 3: Trajectory curvature vs. endpoint error. OT-CFM and SB-CFM produce the straightest trajectories (curvature near 1.0), supporting their low-NFE robustness.

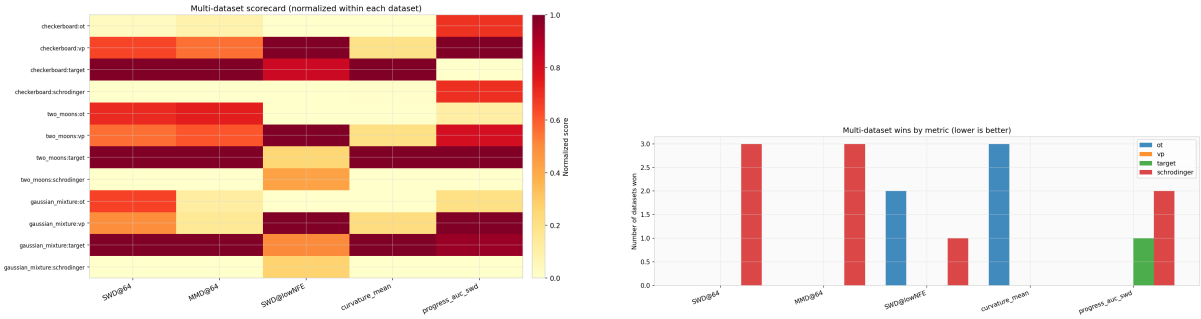
OT-CFM and SB-CFM produce the straightest trajectories (mean curvature close to 1.0), while Target-CFM is substantially more curved. This directly explains the NFE robustness gap: straighter paths require fewer discretization steps to traverse accurately.

4.4 Multi-Dataset Generalization

Extending to three datasets (checkerboard, two_moons, gaussian_mixture) reveals consistent patterns. Table 3 shows win counts across all metrics and datasets.

Table 3: Win counts across three datasets and five evaluation metrics. Each cell shows the number of datasets (out of 3) where the variant achieves the best value.

Variant	SWD@64	MMD@64	SWD@lowNFE	Curvature	AUC
OT-CFM	0	0	**2**	**3**	0
VP-CFM	0	0	0	0	0
Target-CFM	0	0	0	0	1
SB-CFM	**3**	**3**	1	0	**2**



(a) Metric heatmap across datasets.

(b) Win counts by variant and metric.

Figure 4: Multi-dataset comparison. SB-CFM dominates endpoint quality; OT-CFM dominates trajectory straightness and low-NFE robustness.

Time-to-structure. An interesting nuance emerges from the time-to-structure analysis: Target-CFM achieves the best AUC on checkerboard (fastest early structural progress), despite not winning endpoint quality. This suggests that "early geometric progress" and "final sample fidelity" are related but not equivalent objectives.

4.5 Key Takeaways from Study 1

1. **SB-CFM wins endpoint quality** across all three datasets.
2. **OT-CFM wins trajectory straightness and low-NFE robustness** on most datasets.
3. Path design materially affects both sample quality and computational efficiency.
4. All four variants work well in 2D—no instabilities or divergence observed.

The last point will prove critical: the 2D setting masks numerical issues that become catastrophic in high dimensions (Section 6).

5 Study 2: CIFAR-10 Factorial Ablation

5.1 Training Outcomes

Table 4 presents the central result of the CIFAR-10 ablation: only 1 of 8 cells converged to a usable generative model.

Table 4: Training outcomes for all 8 cells. Status indicates whether the training loss converged smoothly, oscillated wildly (unstable), or diverged to NaN. Best FID is the minimum across all 15 solver/NFE combinations. Checkpoint indicates which model was used for evaluation.

Cell	Coupling	Objective	Estimator	Status	Best FID	Checkpoint
000	Independent	CFM	Uniform	NaN	679.2	step 10k
001	Independent	CFM	TPC	NaN	419.8	step 10k
010	Independent	StableVM	Uniform	Unstable	256.8	final
011	Independent	StableVM	TPC	Unstable	197.9	final
100	**BatchOT**	**CFM**	**Uniform**	**Converged**	**18.4**	**final**
101	BatchOT	CFM	TPC	NaN	679.2	step 10k
110	BatchOT	StableVM	Uniform	Unstable	211.4	final
111	BatchOT	StableVM	TPC	Unstable	273.6	final

Immediate observations.

- BatchOT is necessary but not sufficient: cell_100 converged, but cells 101, 110, 111 did not.
- TPC causes NaN divergence in every configuration: cells 001, 101 both diverged (as did cell_000, though for a different reason—see Section 6).
- StableVM causes wild oscillation in every configuration: cells 010, 011, 110, 111 all show severely unstable loss.
- The only successful combination is the simplest improved variant: BatchOT + standard CFM + Uniform sampling.

5.2 Training Loss Curves

Figure 5 shows the training loss evolution for all 8 cells.

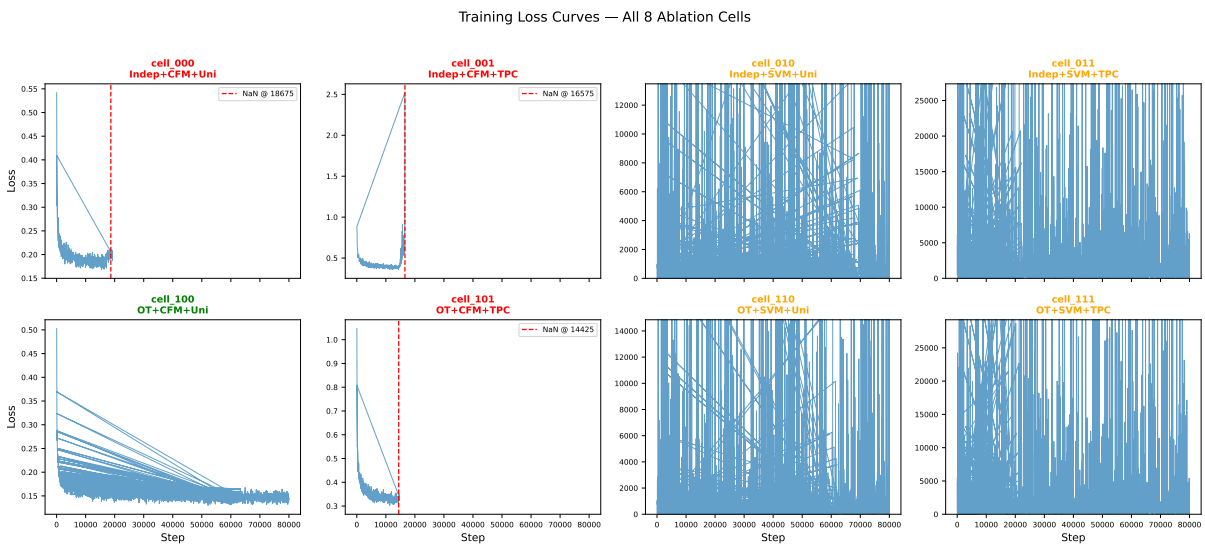


Figure 5: Training loss curves for all 8 ablation cells. Green titles indicate convergence, red indicates NaN divergence, orange indicates unstable oscillation. Red dashed lines mark the step at which NaN first appeared.

Cell_100 (BatchOT + CFM + Uniform) shows a smooth, monotonic decrease from 0.54 to 0.14 over 80k steps, characteristic of healthy flow-matching training.

Cells 000, 001, 101 diverge to NaN at steps ∼18,700, ∼16,600, and ∼14,400 respectively. The pre-NaN loss is relatively low (0.15–0.25), suggesting that the model was learning before numerical instability triggered a cascade.

Cells 010, 011, 110, 111 (all StableVM) exhibit extreme oscillation, with loss swinging between ∼40 and ∼22,000 throughout training. The model never stabilizes, despite 80k steps of training. This pattern is consistent across both coupling types and both estimators, pointing to StableVM as the root cause.

5.3 Cell_100: The Only Successful Model

Since cell_100 is the only cell that produced a meaningful generative model, we analyze it in detail.

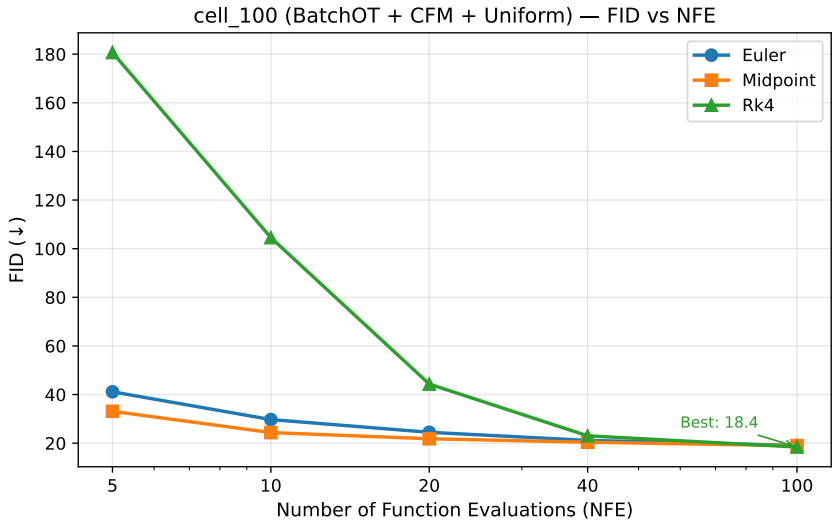


Figure 6: FID vs. NFE for cell_100 (BatchOT + CFM + Uniform), the only converged cell. Three ODE solvers are compared.

Table 5 shows the full FID results for cell_100.

Table 5: FID values for cell_100 across solvers and NFE. Best value in each column is bolded.

Solver	NFE=5	NFE=10	NFE=20	NFE=40	NFE=100
Euler	41.1	29.7	24.5	21.1	19.1
Midpoint	**33.1**	**24.4**	**21.8**	**20.4**	19.0
RK4	180.7	104.6	44.3	23.0	**18.4**

Solver behavior. Midpoint achieves the best FID at low NFE (33.1 at NFE=5), outperforming Euler (41.1) by 19%. RK4 struggles at very low NFE (180.7 at NFE=5) due to its high per-step cost—each "step" actually performs 4 function evaluations, so NFE=5 corresponds to barely one full integration step. At NFE=100, RK4 achieves the best FID of 18.4.

Practical recommendation. For cell_100, midpoint is the most efficient solver: it reaches FID 20.4 at only 40 NFE, matching Euler at 100 NFE. This 2.5× efficiency gain from solver choice is consistent with the trajectory-straightness findings from Study 1—BatchOT coupling produces paths that are well-suited to moderate-order solvers.

5.4 Failed Cells: FID Context

For completeness, Table 7 reports the best FID achieved by each cell, even though only cell_100 represents a meaningful generative model.

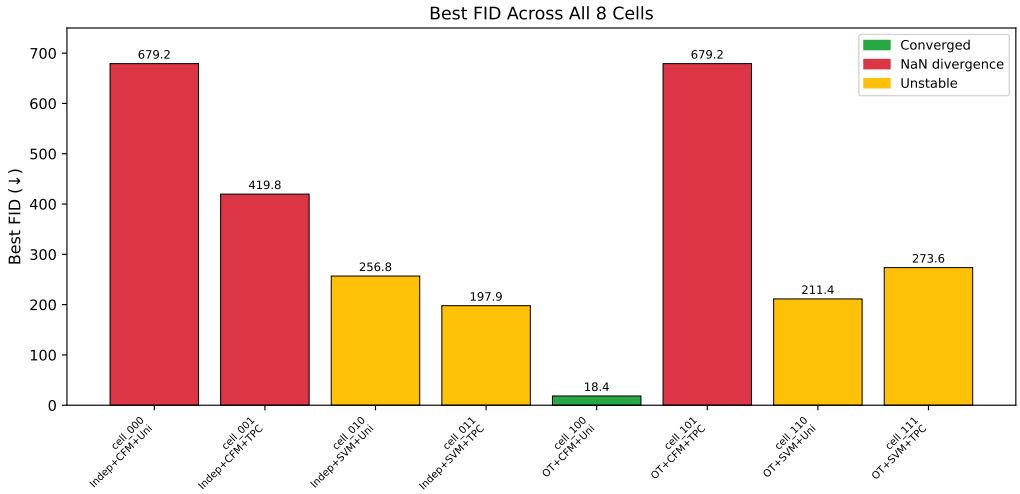


Figure 7: Best FID across all 8 cells. The stark contrast between cell_100 (FID 18.4) and the rest (197–679) underscores the failure of the composition hypothesis.

The FID values of 679.2 for cells 000, 101 correspond to pure noise (the model at step 10k had barely begun learning when evaluated from a pre-NaN checkpoint). Cells 010–011 and 110–111 achieve FIDs in the 197–290 range, reflecting models that learned some coarse structure despite never stabilizing. Cell_001 at 419.8 falls between, having learned slightly more than cells 000/101 before diverging.

6 Discussion

The CIFAR-10 results are overwhelmingly negative for the variance-reduction composition hypothesis. This section analyzes each failure mode, explains the 2D-to-CIFAR gap, and answers our research questions.

6.1 Why Cell_000 (Baseline) Failed

Cell_000 uses the simplest configuration: independent coupling, standard CFM loss, and uniform timestep sampling. It diverged to NaN at step ∼18,700. The root cause is a numerical issue in the OT path's conditional vector field (Eq. 7):

$$u_t(x|x_1) = \frac{x_1 - (1 - \sigma_{\min})x}{1 - (1 - \sigma_{\min})t}. \tag{13}$$

The denominator $1 - (1 - \sigma_{\min})t$ approaches $\sigma_{\min} = 10^{-4}$ as $t \to 1$. Under fp16 mixed precision, this produces values near 10^4, which cascade through the network gradients and eventually overflow.

Why BatchOT prevents this: With independent coupling, the numerator $x_1 - (1 - \sigma_{\min})x$ can be large because x_0 and x_1 are unrelated—a noise sample may be far from its paired data point. BatchOT coupling minimizes the transport cost, keeping $\|x_1 - x_0\|$ small. Since $x_t = tx_1 + \sigma_t x_0$ interpolates between paired points, both the numerator and denominator of u_t remain moderate, preventing the overflow. This explains why cell_100 converges while cell_000 does not: BatchOT is not just a variance-reduction technique—it is a *numerical stabilizer* for the OT path under limited precision.

6.2 Why StableVM Failed

All four StableVM cells (010, 011, 110, 111) exhibit extreme loss oscillation between ∼40 and ∼22,000. The cause lies in the log-probability computation (Eq. 10):

$$\log p_t(x|x_1) = -\frac{d}{2}\log(2\pi) - d\log\sigma_t - \frac{1}{2\sigma_t^2}\|x - \mu_t(x_1)\|^2. \tag{14}$$

The critical term is $-d\log\sigma_t$. For CIFAR-10, $d = 3 \times 32 \times 32 = 3072$. Near $t = 1$, $\sigma_t \approx \sigma_{\min} = 10^{-4}$, so:

$$-d\log\sigma_t \approx -3072 \times \log(10^{-4}) \approx 3072 \times 9.21 \approx 28{,}300. \tag{15}$$

This enormous value dominates the softmax in the importance weights (Eq. 8), causing the weights to concentrate on a single reference sample—whichever $x_1^{(k)}$ is closest to x_t. The resulting target is effectively a single-sample estimate with high variance, but computed through a numerically unstable pathway. The loss oscillation reflects the softmax flipping between different reference samples across training steps.

Why this is invisible in 2D: When $d = 2$, the same term yields $-2 \times \log(10^{-4}) \approx 18.4$, which is small enough that the softmax produces meaningful weight distributions across all K references. The technique works as intended in low dimensions but breaks catastrophically when d is large.

6.3 Why TPC Failed

Cells 001 and 101 diverged to NaN at steps ∼16,600 and ∼14,400 respectively. The TPC consistency penalty (Eq. 12) compares velocity predictions at complementary timesteps:

$$\lambda \left\| v_\theta(x_t^{(1)}, t_1) - v_\theta(x_t^{(2)}, t_2) \right\|^2, \quad t_2 = 1 - t_1. \tag{16}$$

When t_1 is close to 0 and t_2 close to 1, the velocity magnitudes at these two timesteps are fundamentally different: near $t = 0$ the flow must move slowly from noise, while near $t = 1$ it must make fine adjustments toward the data manifold. The conditional vector field magnitudes differ by orders of magnitude due to the $1/(1-(1-\sigma_{\min})t)$ denominator. The consistency penalty produces gradients that are proportional to this mismatch, leading to gradient explosions that destabilize training.

Note: Cell_000 also diverged to NaN without TPC, but for a different reason (fp16 overflow from independent coupling). The NaN onset is earlier for TPC cells (14–17k steps vs. 18.7k), suggesting that TPC accelerates the instability.

6.4 The 2D-to-CIFAR Gap

All four flow-matching variants in Study 1 work perfectly in 2D, with no sign of the catastrophic failures observed on CIFAR-10. The fundamental reason is **dimensionality**:

- StableVM's log-probability scales as $O(d)$: harmless when $d = 2$, catastrophic when $d = 3072$.
- The OT path's denominator creates velocities of magnitude $\sim 1/\sigma_{\min}$. In 2D, the gradient norm stays bounded; in $d = 3072$, the per-channel gradient accumulation amplifies the instability.
- TPC's consistency penalty couples velocities at t and $1 - t$ whose magnitudes diverge more severely in high dimensions, where the velocity field must navigate a higher-dimensional landscape.

This gap carries a methodological warning: **2D experiments are valuable for building geometric intuition but insufficient for validating numerical stability**. Any technique involving log-probabilities, importance weights, or time-endpoint comparisons should be stress-tested in the target dimension before deployment.

6.5 Answering the Research Questions

RQ1: Are the three axes additive? No. The results are emphatically negative. Not only are the axes not additive, but combining BatchOT with either StableVM or TPC actively *destroys* the convergence that BatchOT alone provides. Cell_100 (BatchOT only) converges to FID 18.4; adding TPC (cell_101) causes NaN divergence; adding StableVM (cell_110) causes wild instability.

RQ2: Which single axis provides the largest improvement? BatchOT coupling, unambiguously. It is the only axis whose inclusion leads to convergence (when used alone). StableVM and TPC both degrade performance in every tested configuration. BatchOT acts as both a variance reducer (shorter, straighter trajectories) and a numerical stabilizer (bounded velocity magnitudes).

RQ3: Do 2D insights transfer? Partially. The geometric insight from Study 1—that OT-type paths produce straighter trajectories and better low-NFE robustness—transfers directly: cell_100's midpoint solver at NFE=40 matches Euler at NFE=100, confirming the trajectory-straightness advantage. However, the stability insights do *not* transfer: techniques that work flawlessly in 2D fail catastrophically in high dimensions.

7 Limitations

Several design choices and constraints limit the generalizability of our findings:

Single seed. All CIFAR-10 cells were trained with a single seed (seed 0). While the training failures are consistent and reproducible (all StableVM cells oscillate, all TPC cells diverge), the exact FID of the converged cell_100 carries no error bars. Multi-seed runs would strengthen confidence in the FID=18.4 result.

Fast mode only. Due to compute constraints, all cells used the fast training configuration (9M parameters, 80k steps) rather than the full configuration (47M parameters, 150k steps). While the relative comparisons remain valid (all cells share the same budget), the absolute FID of 18.4 is higher than what a fully-trained model would achieve. The failures, however, are likely to persist or worsen with longer training, since the instabilities appear early ($<$ 20k steps).

No gradient clipping. Standard practice in diffusion/flow-matching training includes gradient norm clipping (typically to 1.0). Our training pipeline does not use gradient clipping, which likely exacerbates the NaN divergence in cells 000, 001, and 101. With gradient clipping, some of these cells *might* survive longer, though the underlying numerical issues would remain.

fp16 mixed precision. All training used fp16 automatic mixed precision. The OT path denominator $1 - (1 - \sigma_{\min})t$ reaches 10^{-4} near $t = 1$, which is within fp16's representable range but close to its precision limit. Training in fp32 might prevent some NaN events, though at $2\times$ memory cost.

Different codebases. Study 1 and Study 2 use independent codebases with different network architectures (MLP vs. U-Net), optimizers (lr=10^{-3} vs. 10^{-4}), and data domains (2D vs. images). While this mirrors realistic research settings, it means that direct numerical comparisons between the two studies are not meaningful—only qualitative insights transfer.

CIFAR-10 only. Results on CIFAR-10 (32×32) may not generalize to higher-resolution datasets. The dimension-dependent failures of StableVM would only worsen at higher resolutions (e.g., $d = 3 \times 256 \times 256 = 196{,}608$ for 256×256 images).

8 Conclusion

We investigated whether three independently proposed variance-reduction techniques for Conditional Flow Matching—BatchOT coupling, StableVM objective, and TPC estimator—compose additively when combined. Through a 2^3 factorial ablation on CIFAR-10, complemented by 2D synthetic experiments, we find that the answer is decisively negative.

BatchOT coupling is the single most impactful technique. It is the only axis whose inclusion leads to successful training (cell_100, FID 18.4), acting simultaneously as a variance reducer and a numerical stabilizer for the OT probability path. This finding is consistent with **?** and validates BatchOT as the recommended default for CFM training.

StableVM and TPC fail in high dimensions. StableVM's importance weights scale with the data dimension d, causing softmax collapse when $d = 3072$. TPC's consistency penalty couples velocity predictions at complementary timesteps whose magnitudes differ by orders of magnitude. Both techniques work in 2D ($d = 2$) but break catastrophically on CIFAR-10.

2D experiments inform geometry but not stability. Study 1's finding that OT-type paths produce straighter trajectories transfers to CIFAR-10 (cell_100's solver efficiency confirms this). However, the absence of instabilities in 2D gives false confidence about numerical robustness in high dimensions.

Future directions. The negative results suggest concrete remediation strategies:

- **Dimension-normalized StableVM**: Replace $\log p_t$ with $\frac{1}{d} \log p_t$ in the importance weights to prevent softmax collapse.
- **Gradient-stopped TPC**: Detach the consistency penalty from the computational graph to prevent gradient explosions, or restrict antithetic pairs to $t \in [0.1, 0.9]$ to avoid extreme velocity mismatches.
- **Gradient clipping + fp32**: Standard numerical safeguards may allow the baseline (cell_000) and TPC cells to converge.
- **Multi-seed validation**: Confirm the cell_100 result with 3–5 seeds to establish confidence intervals.

These results carry a broader methodological lesson: variance-reduction techniques should be validated not only for their statistical properties but also for their numerical stability in the target dimension. A technique that reduces variance in expectation but introduces numerical catastrophes in practice is worse than the baseline it aims to improve.